Veer Kakar

veerkakar1@gmail.com | linkedin.com/in/VeerKakar | github.com/VeerKakar17 | PortfolioWebsite

Education

University of Pennsylvania Philadelphia, PA
BSE in Computer Science | GPA: 3.99/4.0 *Aug. 2024 – May 2028*

University of Pennsylvania Philadelphia, PA
MSE in Computer Science, Systems Concentration | GPA: 4.0/4.0 *Aug. 2026 – May 2028*
Relevant Coursework: Distributed Systems, GPU Programming, Databases, Embedded Systems, Network Security

Experience

Software Engineering Intern May 2026 – Present
Roblox *San Mateo, CA*
- Owned 2 IP licensing initiatives end to end, authoring 3 specs and post-launch metrics analysis.
- Led 10+ cross-team syncs across platform, data science, product, and design to set direction.
- Defined service boundaries, schemas, RPC contracts, consistency behavior, and asynchronous execution; built durable Temporal workflows with persisted state and automatic recovery from partial failures.
- Launched a production feature that removed manual content inspection, saving IP holders 2+ hours weekly.
- Architected a cross-service feedback signal, resolving a months-long backlog clutter pain point.

Software Engineering Intern May 2025 – August 2025
Transforma Tech, Inc - an IBM Business Partner *Exton, PA*
- Designed & developed a scalable Java application for automatically generating IBM App Connect Enterprise applications, significantly accelerating production and enabling faster client-specific customization.
- Integrated enterprise applications for clients (Humana, SMBC), ensuring scalable data pipelines & reliable systems.
- Improved throughput with asynchronous data exchange between heterogeneous systems using IBM MQ and Kafka.

Operating Systems and Data Structures & Algorithms Teaching Assistant May 2025 – Present
University of Pennsylvania *Philadelphia, PA*
- Selected to concurrently TA Operating Systems and DSA, supporting 350+ students across both courses.
- Lead weekly 20-student DSA sections and 150+ student OS recitations; hold course-wide office hours.
- Develop and grade exams, proofs, programming assignments, and systems projects; conduct oral exams and evaluate project demos through system-design questioning.

Associate App Developer Intern May 2024 – August 2024
Aetna Inc, a CVS Health Company *Blue Bell, PA*

Projects

Bare-Metal RPI5 Operating System | *C, AArch64 ASM, ARMv8-A, QEMU, Unix/POSIX* May 2026 – Present
- Built a bare-metal AArch64 Unix-style OS for RPi 5/QEMU with user/kernel isolation and tabbed GUI TTYs.
- Reverse-engineered sparse RPi 5 docs via Linux source to implement memory-mapped drivers and hardware I/O.
- Implemented per-process virtual memory with demand paging, lazy allocation, mmap, and copy-on-write fork.
- Built ext2-style inode VFS and ELF loader; added POSIX signals, pipes, shell, threads, and synchronization.

Mini Minecraft | *C++, Multithreading, OpenGL, Real-Time Rendering, Optimization* February 2025 – May 2025
- Developed a high-performance Minecraft recreation in C++/OpenGL with a focus on systems-level optimization.
- Implemented multithreaded terrain generation and face-culling to reduce rendering overhead by over 60%.
- Designed custom redstone logic system, custom block VBOs, flowing water, and shadow mapping from scratch.

Penn Mobile Android Developer | *Kotlin, Android, CI/CD, GitHub, Team Development* January 2026 – Present
- Collaborate with Design and PMs to maintain the Penn Mobile Android app, with 5k+ downloads.
- Migrated core features from deprecated XML Data Binding to Jetpack Compose; improved performance by 15%.

Technical Skills

Languages: C, C++, Rust, CUDA C++, Python, Java, AArch64 Assembly, SQL, Typescript, C#, Kotlin, OCaml
Systems & Storage: Linux/Unix, Virtual Memory, Filesystems, POSIX, IPC, PostgreSQL, MongoDB, S3
Concurrency & Distributed: Multithreading, Async Workflows, RPCs, Fault Tolerance, Kafka, IBM MQ, Temporal
GPU & Tools: GPU Programming, Performance Optimization, OpenGL, Vulkan, Docker, Git, AWS, RPi5, CI/CD
Interests: Crocheting, Jazz Drumming, Tennis, Taekwondo (3rd degree black belt)